SANGAVARSHINI B

Embedded System & IoT Developer

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Objective

Dedicated and enthusiastic Embedded Systems and IoT Developer with knowledge in ESP32, Arduino, sensors, and cloud-based IoT applications. Skilled in Python, C, and MicroPython with hands-on experience in MQTT communication, ThingSpeak, PCB design, and IoT monitoring systems. Quick learner with strong analytical thinking, teamwork, and problem-solving abilities.

Education Qualifications

B.E – Electronics and Communication Engineering **2022 – 2026**

Thanthai Periyar Government Institute of Technology, Vellore. | CGPA: 8.07 (80%)

HSC 12th **2021 – 2022**

Sri Venkateswara Matric Hr. Sec. School, Pappireddipatti. | 90.16%

SSLC 10th **2019 – 2020**

Govt. Girls Hr. Sec. School, Pappireddipatti. | 86.66%

Project

Energy-Efficient Low-Power IoT Devices Using Deep Learning

Low-power IoT system using ESP32 with deep learning-driven sleep/wake scheduling for optimized sensor data transmission and cloud monitoring through ThingSpeak.
Stack: ESP32, Deep Learning, MQTT, ThingSpeak, Python, IoT Sensors.

Air Quality Monitoring System

Real-time environmental monitoring system using gas and environmental sensors with wireless cloud-based visualization and pollution alert generation.
Stack: ESP32, MQ Sensors, DHT11/DHT22, ThingSpeak, Wi-Fi, IoT.

Underwater Wireless Communication System

Underwater communication system using acoustic and RF signal transmission for reliable marine monitoring and underwater data exchange in challenging environments.
Stack: Acoustic Communication, RF Modules, Embedded Systems, Sensors, Signal Processing.

AR Application using Unity

Augmented Reality visualization tool integrated with IoT sensor data for interactive system demonstration.

Stack: Unity, C, IoT APIs.

Skills

Programming Languages
- C Programming
- Python
- MicroPython

Embedded Systemn & IoT
- Embedded System
- IoT
- ESP32
- Arduino

Tools & Technologies
- MATLAB
- Power BI
- PCB Design
- Unity (AR Application)

IoT Protocols & Cloud
- MQTT Protocol
- ThingSpeak Cloud
- Cloud IoT

Soft Skills
- Innovative and Analytical Thinking
- Quick Learner with Practical Approach
- Strong Teamwork and Adaptability

Communication Skills

- Tamil - Native
- English - Fluent

Certifications

- Awarded 3rd Place in the 'TNWISE Women's Hackathon'
- Awarded 2nd Place in the 'National Level Technical Symposium – VOLTERA' 26'
- Presented a Energy-Efficient Low-Power IoT Devices Using Deep Learning project at a National Conference (RTPEPS'26)
- Participated in 'Watt A Presentation', showcasing technical presentation and communication skills
- Participated in 'Innovision – Aatral'25', demonstrating interest in emerging technologies and innovation

Internship Experience

Zeekers Technology Solutions Pvt. Ltd, Coimbatore – 641 006.

IoT & Embedded Systems Intern — July 2025 – Aug 2025.

- Worked on IoT development projects using Embedded C, ESP32, and sensor interfacing techniques.
- Gained hands-on experience in embedded systems, interfacing, and wireless communication.